



Subject: WCN Lab

Class: TE-Data Science

Semester: V

A.Y. 2025-2026

Experiment No.6

1. Aim: Construct front end applications using React and back end using Node.js/express.
2. Software used: node js/express, react
3. Theory :- Problem

Statement :

Design and implement a simple full-stack web application where a React.js frontend communicates with a Node.js + Express backend through HTTP requests.

The application should demonstrate the request–response cycle using:

1. A GET request to fetch a welcome message from the backend and display it in the React frontend.
2. A POST request where the frontend sends sample JSON data (e.g., { name: "RAM", age: 25 }) to the backend. The backend should log the received data and return a response confirming receipt, which should be displayed in the React frontend.

In modern web applications, the frontend (client) and backend (server) work together to deliver dynamic functionality.

- Frontend (React):
The frontend is the user interface that runs in the browser. It is responsible for collecting input from users and displaying information received from the backend. React provides a declarative way to build UI components and supports asynchronous communication with servers using fetch() or axios.

- Backend (Node.js + Express):
The backend handles data processing and business logic. Using Express.js, developers can create RESTful APIs that respond to client requests.
 - A GET request is typically used to retrieve data from the server.
 - A POST request is used to send data to the server, usually in JSON format.
- Communication Process (Request–Response Cycle):
 1. The React frontend makes a request (GET/POST) to the Node.js backend.
 2. The backend receives the request, processes it, and sends a response.
 3. The frontend displays the response to the user.
- CORS (Cross-Origin Resource Sharing):
Since React (running on port 3000) and Node.js (running on port 5000) are different origins, the backend must enable CORS to allow communication between them. This is done using the cors middleware in Express.
- JSON Data Exchange:
The common format for sending and receiving data is JSON (JavaScript Object Notation). In this experiment, the frontend sends:
- `{ "name": "RAM", "age": 25 }`

The backend logs this data and responds with a confirmation JSON.

Conclusion:

Constructed front end applications using React and back end using Node.js/express.

